Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

CS3230

Tutorial

Week 10

Question

Suppose Bob has a collection of music files that he wants to burn into CDs. Each CD has a storage capacity of 100MB. In addition, Bob does not want to store more than two music files per CD. Note that each music file cannot be split and hence cannot be burned on more than 1 CDs. Derive the algorithm for obtaining the the minimum number of CDs that Bob needs to burn his music files on the discs

Input: n integers $a_1, a_2, \ldots, a_n$ denoting the sizes of n files.